PHYSICS PRACTICAL SHEETS

KV Campus

Date: 11 June 2026
Clsss: CE - I/II
Roll No.: 43
Shift: Morning
Experiment No.: 2
Group: S
Sub.: Physics lab 102
Set: I

Object of the Experiment (Block Letter)

MEASUREMENT OF MOMENT OF INERTIA OF FLYWHEEL

Apparatus required:

1) Flywheel
2) Stopwatch
3) Meter Scale
4) Thread
5) weight hanger
6) Known Weights
7) Marker
8) Vernier Calipers

Theory:

When a mass m, attached to the axle of a flywheel via a thread, falls through a height h, its potential energy is transferred into:

a) Translational kinetic energy of the mass
b) Rotational kinetic energy of the flywheel
c) Work done against friction

Using the principles of energy conservation and accounting for the work done against friction, the moment of inertia (I) of the flywheel can be determined using the following formula:

$$I = \frac{2mgh - mr^2\omega^2}{\omega^2\left(1 + \frac{n}{n_1}\right)} \quad \text{---} \quad (1)$$

where,

r is the radius of the axle

n is number of turns of the thread wound around the axle

n_1 is the number of revolutions made by the flywheel before coming to rest after the string is detached.

ω is the angular velocity of the flywheel

$$\omega = 2 \times \frac{2\pi n_1}{t} \quad \text{——} \quad (2)$$

h is the vertical distance through which the mass m falls as the string unwinds from axle of the flywheel

Observations:

Vernier constant = 0.1 mm = 0.01 cm

Diameter of the axle, d = (i) 2.14 cm (ii) 2.16 cm (iii) 2.15 cm

Mean diameter (d) = 2.15 cm

Radius of the axle, r = 1.075 cm

Circumference of the wheel = 58 cm

For mass: (i) M = 200g, Height h = 71 cm

No. of turns wound an axle, n = 10

(ii) m = 250g, height h = 70.5 cm

No. of turns wound an axle, n = 10

(iii) m = 300g, height h = 71 cm

No. of turns wound an axle, n = 10

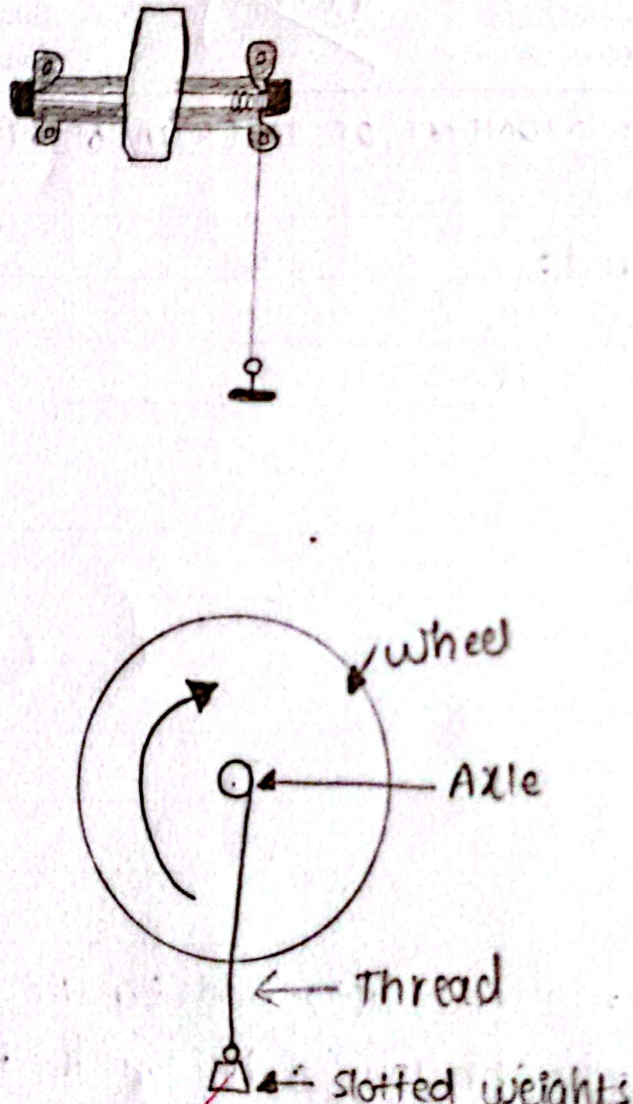

Fig 1: Flywheel mounted on the wall

Table 1: Determination of the moment of inertia of the flywheel

Mass m	SN	No of complete revs (made by the wheel) x	Fraction of revolution	Total No. of revs $(x+y)$	Mean n_1	Time	Mean time t	Angular velocity (ω) $\omega = \frac{4\pi n_1}{t}$	Moment of inertia $I = \frac{2mgh - mr^2\omega^2}{\omega^2(1+\frac{n}{n_2})}$
	1	14	0.71	14.71		27			
200 gm	2	14	0.69	14.69	14.74	28	26.67	6.94	3.44×10^5
	3	14	0.82	14.82		25			
250 gm	1	20	0.09	20.09	19.58	30	30.3	8.12	3.46×10^5
	2	19	0.01	19.01		28			
	3	19	0.56	19.56		33			
300 gm	1	22	0.76	22.76	23.12	30	31	9.37	3.31×10^5
	2	23	0.77	23.77		32			
	3	22	0.83	22.83		31			

Mean moment of inertia, $I = 3.4 \times 10^5$ gm cm^2

Result:

The moment of Inertia (I) of the flywheel is found to be 3.4×10^5 gm cm^2

Precautions:

1) Minimize friction in the flywheel and its axle to ensure accurate measurements.

2) Check that the string is free of kinks or twists to avoid uneven rotation.

3) Use a thin, strong string & wind it evenly around the axle for consistent results.

4) start the stopwatch precisely when the mass detaches from the axle to record accurate timing.

Conclusion: The moment of inertia of the flywheel is determined from the experiment.

11th June 2026.